

Searching Text Strings from Files Using Python

Searching text strings from files in a given folder is easily accomplished by using Python in Windows. While Linux has the *grep* command, Windows does not have an equivalent. The only alternative, then, is to make a command that will search the string. This article introduces *see.py*, which helps in accomplishing this task.



Have you ever thought of searching a string in the files of a given folder? If you are a Linux lover, you must be thinking about the *grep* command. But in Windows, there is no *grep* command. By using Python programming, you can make your own command which will search the string pattern from the given files. The program also offers you the power of regular expressions to search the pattern.

In this article, I am going to show you an amazing utility, which will help you to find the string from a number of files.

The program, *see.py*, will search for the string pattern provided by the user, from the files presented in the directory, also given by the user. This is equivalent to the *grep* command in the Linux OS. Here, we will use Python 2.7.

The program expects the string pattern and directory from the user. Let us examine the code and discuss it.

```

import the mandatory modules
import os
import re
import sys
import argparse

```

In the following code I have declared a class *Text_search* class *Text_search* :

```

def __init__(self, string2, path1,i=None):

```

```

    self.path1= path1
    self.string1= string2
    self.i=1
    if self.i:
        string2= string2.lower()
        self.string2= re.compile(string2)

```

The following method gives the file's name in which the given string is found:

```

def txt_search(self):
    file_number = 0
    files = [f for f in os.listdir(self.path1) if os.path.
isfile(self.path1+"/"+f)]
    for file in files:
        file_t = open(self.path1+"/"+file)
        file_text= file_t.read()
        if self.i:
            file_text=file_text.lower()
        file_t.close()
        if re.search(self.string2, file_text):
            print "The text "+self.string1+" found in ",
file

        file_number=file_number+1
    print "total files are ",file_number

```